Active Directory Red-Team & Blue-Team Home Lab

Controlled Ransomware Simulation Report

Author: Yatish Waghmare

MSc Cyber Security (Infrastructures Security)

November 13, 2025

1 Executive Summary

This report describes a self-built cybersecurity home lab created to simulate a complete ransomware attack and defense scenario inside an isolated Active Directory environment. The goal was to understand the entire attack lifecycle, from discovery and credential compromise to detection, containment, and recovery, using enterprise-grade tools in a safe and controlled setup.

The red team phase focused on reproducing the behavior of a real attacker by performing network reconnaissance, password spraying, and remote code execution with Impacket PsExec. A PowerShell script was executed to imitate ransomware activity by encrypting test files and creating a ransom note for analysis. The blue team phase concentrated on monitoring and response using Splunk. Windows event logs were collected through the Universal Forwarder and correlated to detect failed logons, suspicious PowerShell activity, and sudden bursts of file changes. During the simulation, more than 300 file access events were captured, providing a clear timeline of the attack and allowing detailed forensic analysis.

This project demonstrated that with proper monitoring, well-tuned detections, and a planned recovery process, even a fast-moving ransomware attack can be identified and contained before serious damage occurs. It also strengthened my skills in Active Directory security, Splunk detection engineering, PowerShell automation, and incident response.

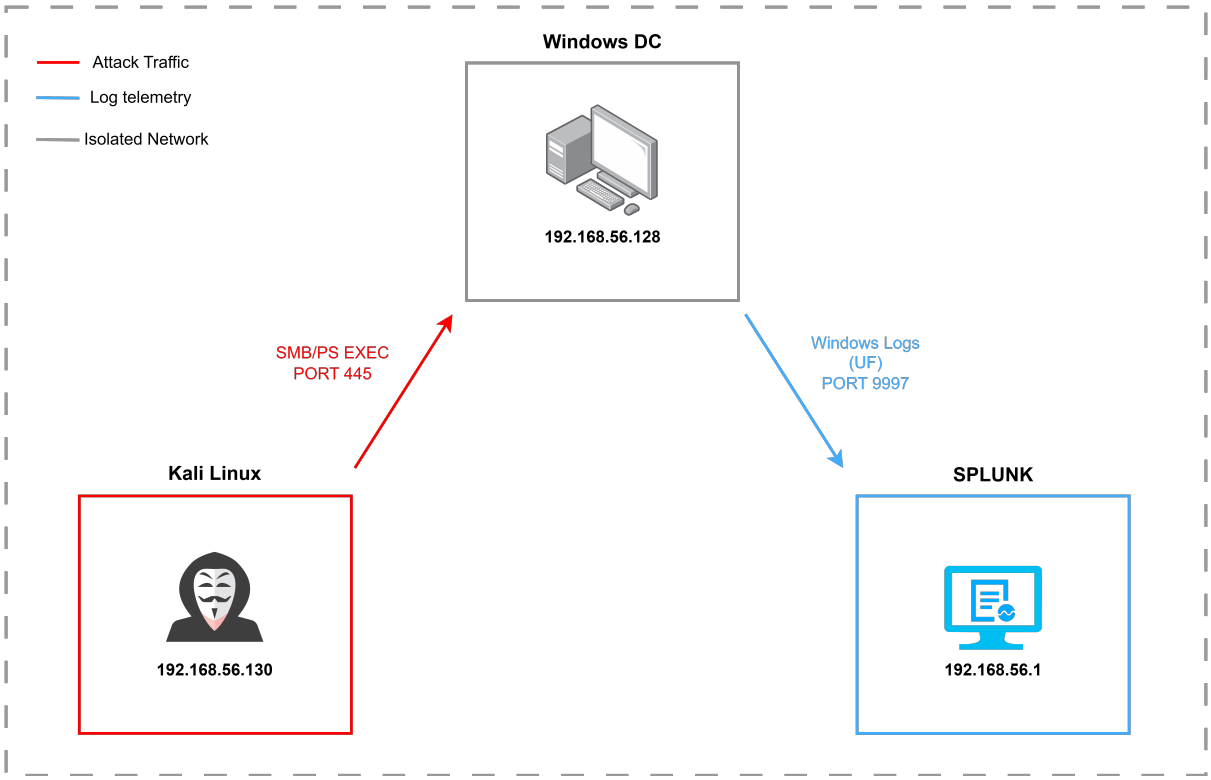
All actions were performed safely within an isolated lab, and every system was fully restored after the test. The work reflects my dedication to practical cybersecurity, a strong understanding of both offensive and defensive operations, and readiness for professional roles in security consulting or SOC analysis.

2 Lab Overview

2.1 Environment Setup

- **Attacker:** Kali Linux VM (Impacket, nmap, Python HTTP server)
- **Target:** Windows Server 2022 Domain Controller (DC1)
- **Monitoring:** Splunk Enterprise with Universal Forwarder
- **Network:** Isolated virtual subnet (192.168.56.0/24)

2.2 Diagram



3 Red Team Simulation

3.1 Objective

Simulate credential abuse and ransomware-style file encryption within a safe, isolated Active Directory environment.

3.2 Methodology

- Performed network discovery and enumeration using `nmap`.
- Conducted password spraying to identify weak domain credentials.
- Gained remote execution via Impacket `psexec`.
- Executed a PowerShell script to simulate ransomware behaviour: renaming files and creating a ransom note.

3.3 Tools Used

Kali Linux, Impacket, PowerShell, nmap, Python HTTP Server.

3.4 Observation

The attack successfully simulated a controlled ransomware event, producing observable file encryption behaviour within the lab environment. All activities were contained and safely reverted using recovery scripts.

4 Blue Team Detection & Response

4.1 Detection Engineering

- Deployed Splunk Universal Forwarder to collect Windows Security logs.
- Monitored key Event IDs: 4624/4625 (Authentication), 4663 (File Access), 5156 (Network Connection).
- Built correlation searches to detect credential abuse, suspicious PowerShell downloads, and rapid file modifications.

4.2 Evidence

Splunk captured over 300 file access and write events during the simulated encryption stage, providing a complete forensic timeline.

4.3 Response

Recovery was validated through execution of an automated PowerShell script (`revert.ps1`), restoring all encrypted files.

5 Outcome

During the ransomware simulation, approximately 300 file access and write events were recorded in the target directory, revealing how quickly an attacker could encrypt or damage files if left undetected. The activity was safely contained in my isolated home lab, and all files were restored afterward.

Using Splunk telemetry, I developed detection rules correlating failed logons, unusual PowerShell activity, and spikes in file modifications. These detections fed into a response playbook that automatically isolates affected hosts and initiates recovery. The exercise clearly demonstrated both the potential impact of ransomware and the effectiveness of timely monitoring and response.

6 Skills Gained

- Active Directory Security
- Splunk Detection Engineering
- PowerShell Automation
- Impacket Toolkit
- Nmap
- Incident Detection and Response
- Windows Event Log Analysis
- Forensic Investigation
- Threat Simulation
- Security Monitoring

7 Key Takeaways

- Learned how quickly ransomware can escalate and how telemetry timing affects detection.
- Gained experience correlating multiple event sources for reliable alerting.
- Strengthened ability to translate offensive activity into defensive detections.
- Reinforced the importance of isolated, safe lab practices for research.

8 Recommendations

- Enforce strong password policies and implement MFA for privileged accounts.
- Restrict admin shares and limit PowerShell egress.
- Maintain offline backups and regularly test restore procedures.
- Implement Splunk alerts for authentication anomalies and rapid file changes.

9 Conclusion

This project demonstrated a full attack–detection–recovery lifecycle in a controlled environment. The experience reinforced real-world skills in Active Directory security, Splunk detection engineering, and incident response. It also demonstrated the value of proactive monitoring and automation in mitigating ransomware risks. This lab reflects my initiative, technical depth, and readiness for professional roles in cybersecurity consulting or SOC operations.

All experiments were conducted safely in an isolated environment. No external systems were affected.